

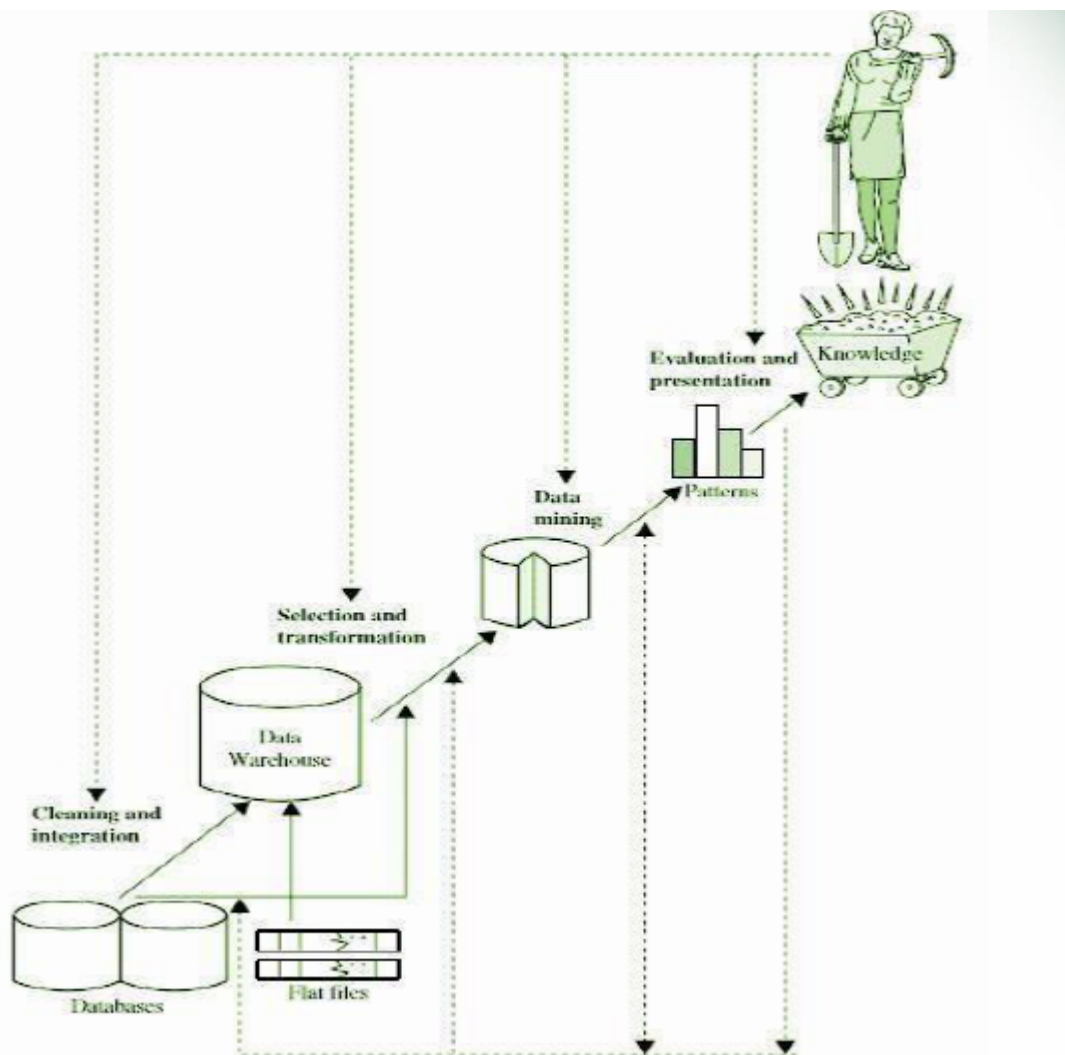
What is Data Mining?

Data mining is the process of **searching for useful knowledge or interesting patterns** from large volumes of data.

It involves identifying hidden relationships, trends, and rules that are not immediately visible.

Data mining is a **core and essential step** in the broader process known as **Knowledge Discovery in Databases (KDD)**.

Knowledge Discovery Process (KDD)



The **Knowledge Discovery Process** is an **iterative sequence of steps** used to convert raw data into meaningful knowledge.

1. Data Cleaning

Data cleaning involves **removing noise and correcting inconsistent data**.

- Handles missing values
- Removes duplicate records
- Corrects errors and outliers

Purpose:

To improve data quality and reliability.

2. Data Integration

Data integration is the process of **combining data from multiple data sources** into a single dataset.

- Merges databases, files, or data streams
- Resolves conflicts among datasets

Purpose:

To provide a unified and complete view of data.

3. Data Selection

Data selection involves **retrieving only relevant data** required for the mining task.

- Filters unnecessary attributes
- Selects appropriate records

Purpose:

To reduce data volume and focus on useful information.

4. Data Transformation

Data transformation converts data into a **form suitable for data mining**.

- Aggregation
- Normalization
- Feature construction
- Data consolidation

Purpose:

To improve mining efficiency and accuracy.

5. Data Mining

This step applies **data mining algorithms** to extract patterns.

- Classification
- Clustering
- Association rule mining
- Regression

Purpose:

To discover meaningful patterns and relationships.

6. Pattern Evaluation

Pattern evaluation identifies **interesting and useful patterns**.

- Removes redundant or irrelevant patterns
- Measures pattern significance

Purpose:

To ensure only valuable knowledge is retained.

7. Knowledge Presentation

This step presents discovered knowledge in an **understandable form**.

- Charts, graphs, and reports
- Rules and visual summaries

Purpose:

To support decision-making.

What kinds of data can be mined ?

The most basic forms of data for mining applications are database data, data warehouse data and transactional data. Data mining can also be applied to other forms of data such as data streams, ordered/sequence data, graph or networked data, spatial data, text data, multimedia data and the WWW.

1. Database Data:

- Traditional structured data stored in relational databases.
- Examples: Customer records, sales transactions, employee information.

2. Data Warehouse Data:

- Consolidated and historical data from multiple sources.
- Often used for analytical purposes.

- Examples: Aggregated sales data over several years, inventory trends.

3. Transactional Data:

- Data generated from business transactions.
- Examples: Purchase histories, online orders, ATM withdrawals.

4. Data Streams:

- Continuous flow of data that arrives in real-time.
- Examples: Sensor readings, stock market feeds, social media streams.

5. Ordered/Sequence Data:

- Data where the order of items matters.
- Examples: DNA sequences, web clickstream data, time-series data.

6. Graph or Networked Data:

- Data represented as nodes and edges showing relationships.
- Examples: Social networks, communication networks, citation networks.

7. Spatial Data:

- Data related to geographic locations and coordinates.
- Examples: GPS data, satellite images, city maps.

8. Text Data:

- Unstructured or semi-structured textual information.
- Examples: Emails, reports, product reviews, web pages.

9. Multimedia Data:

- Data in forms like images, audio, and video.
- Examples: Medical imaging, videos, audio recordings.

10. Web Data (WWW):

- Data from websites and online platforms.
- Examples: Clickstream data, web page content, user interactions.

What kinds of patterns can be mined?

- Class/Concept Description: Characterization and Discrimination
- Mining Frequent Patterns, Associations, and Correlations
- Classification and Regression for Predictive Analysis
- Cluster Analysis
- Outlier Analysis

1. Class/Concept Description (Characterization and Discrimination)

- **Characterization:** Summarizes general features of a target class of data.
 - Example: “Most customers buying laptops are aged 20–35 and prefer online purchases.”
- **Discrimination:** Compares characteristics of different classes to distinguish them.
 - Example: “Customers buying desktops are older than those buying laptops.”

2. Mining Frequent Patterns, Associations, and Correlations

- **Frequent Patterns:** Items or events that occur together often.

- Example: “Customers who buy bread often buy butter.”
- **Association Rules:** Relationships between items in a dataset.
 - Example: If a customer buys a smartphone, they are likely to buy a protective case.
- **Correlation Analysis:** Measures the strength of relationships between variables.
 - Example: “Higher income correlates with higher spending on luxury products.”

3. Classification and Regression (Predictive Analysis)

- **Classification:** Assigns data to predefined categories.
 - Example: “Email is classified as spam or not spam.”
- **Regression:** Predicts a continuous value based on input data.
 - Example: “Predict house prices based on area, location, and number of rooms.”

4. Cluster Analysis

- Groups similar data points together without predefined categories.
 - Example: “Segment customers into groups based on purchasing behavior.”
- Useful for discovering hidden structures or patterns in data.

5. Outlier Analysis

- Identifies rare or abnormal data points that deviate from the norm.
 - Example: “Detect fraudulent credit card transactions or abnormal sensor readings.”
- Helps in fraud detection, error identification, and unusual event detection.

Technologies Used in Data Mining

Statistics

- Provides mathematical foundation for analyzing and summarizing data.
- Used for:
 - Data summarization (mean, median, mode)
 - Probability and modeling
 - Hypothesis testing
 - Correlation and regression
 - Outlier detection
- **Example:** Predicting customer purchase behavior using regression analysis.

Machine Learning

- Allows systems to **learn patterns from data** and make predictions or decisions.
- Types:
 - **Supervised Learning:** Classification and regression (e.g., spam detection)
 - **Unsupervised Learning:** Clustering and association (e.g., market segmentation)
 - **Reinforcement Learning:** Learning by trial and error (e.g., recommendation engines)

Database Systems / Data Warehousing

- Provide efficient storage, retrieval, and management of large volumes of data.
- **Database Technology:** Handles structured data and supports queries.
- **Data Warehousing:** Integrates data from multiple sources for analysis.
- **Example:** A retail chain storing sales data from all stores in a warehouse for trend analysis.

Artificial Intelligence (AI)

- AI techniques help **model complex patterns and perform intelligent reasoning.**
- Techniques used:
 - Expert systems
 - Neural networks
 - Fuzzy logic
- **Example:** Fraud detection using neural networks.

Visualization Technology

- Enables the **graphical representation of data and patterns** for better understanding.
- Techniques:
 - Charts, graphs, heatmaps, dashboards
 - Interactive visualizations for large datasets
- **Example:** Visualizing customer clusters in a 3D scatter plot.

Neural Networks

- Computational models inspired by the human brain.

- Used for pattern recognition, classification, and prediction in **complex datasets**.
- **Example:** Handwriting recognition, image classification.

Online Analytical Processing (OLAP)

- Enables **interactive analysis of multidimensional data**.
- Operations: Roll-up, drill-down, slicing, dicing
- **Example:** Analyzing sales by region, time, and product category.

Text Mining and Web Mining

- **Text Mining:** Extracts patterns and knowledge from unstructured text data.
 - Example: Sentiment analysis of customer reviews.
- **Web Mining:** Extracts patterns from web data (clickstreams, web pages, logs).
 - Example: Recommendation systems based on browsing behavior.

What is a Data Warehouse?

A **data warehouse** is a **centralized repository** that stores **large volumes of historical data** collected from multiple sources.

It is mainly used for **analysis, reporting, and decision-making**, not for daily transactions.

Key Characteristics of a Data Warehouse

1. **Subject-Oriented**

- Organized around key subjects such as sales, customers, or products.

2. **Integrated**

- Data from different sources is combined into a consistent format.

3. **Time-Variant**

- Stores historical data over long periods (years).

4. **Non-Volatile**

- Data is stable; once entered, it is not frequently updated or deleted.

Components of a Data Warehouse

1. **Data Sources**

- Databases, files, web data, spreadsheets, etc.

2. **ETL Process (Extract, Transform, Load)**

- **Extract:** Collect data from sources
- **Transform:** Clean and convert data
- **Load:** Store data in the warehouse

3. **Data Warehouse Storage**

- Central database storing processed data.

4. **OLAP Tools**

- Used for analysis (roll-up, drill-down, slice, dice).

5. **Front-End Tools**

- Reporting, dashboards, and data mining tools.

Schema Types

1. **Star Schema**

Definition

A **Star Schema** is a data warehouse schema where a **central fact table** is connected directly to multiple **dimension tables**, forming a **star-like structure**.

Structure

- **Fact Table**
 - Contains **quantitative measures** (facts)
 - Example: Sales_Amount, Quantity, Profit
 - Includes **foreign keys** linking to dimension tables
- **Dimension Tables**
 - Contain **descriptive attributes**
 - Example: Time, Product, Customer, Location
 - Each dimension table is **denormalized**

2. Snowflake Schema

Definition

A **Snowflake Schema** is an extension of the star schema where **dimension tables are normalized** into multiple related tables, forming a **snowflake-like structure**.

Structure

- **Fact Table**
 - Same as star schema

- **Dimension Tables**

- Split into multiple normalized tables
- Reduces data redundancy

Database	Data Warehouse
Used for transactions	Used for analysis
Current data	Historical data
Frequent updates	Mostly read-only
Application-oriented	Decision-oriented

Applications of Data Warehousing

- Business intelligence (BI)
- Sales and marketing analysis
- Financial reporting
- Trend prediction
- Data mining

Advantages

- Faster query performance
- Better decision-making
- Integrated view of data
- Supports data mining and analytics

Basic statistical descriptions of Data.

Basic statistical descriptions help us **summarize, understand, and interpret data** clearly before performing advanced analysis or data mining.

1. Measures of Central Tendency

These describe the **center or typical value** of a dataset.

- **Mean:** Average value of data
- **Median:** Middle value of ordered data
- **Mode:** Most frequently occurring value

✦ *Purpose:* To know the general tendency of data.

2. Measures of Dispersion (Variability)

These describe how **spread out** the data values are.

- **Range:** Difference between maximum and minimum values
- **Variance:** Average of squared deviations from the mean
- **Standard Deviation:** Square root of variance

✦ *Purpose:* To understand data consistency and variability.

3. Measures of Position

These show the **relative position** of values in a dataset.

- **Quartiles (Q1, Q2, Q3)**
- **Percentiles**
- **Deciles**

✦ *Purpose:* To compare individual data values within a dataset.

4. Shape of Data Distribution

These describe how data is distributed.

- **Symmetric distribution**
- **Skewed distribution (positive or negative)**
- **Kurtosis:** Peakedness or flatness of data

✦ *Purpose:* To identify data behavior and suitability of statistical methods.

5. Graphical Statistical Descriptions

Visual representations that support numerical summaries.

- **Histogram**
- **Bar chart**
- **Box plot**
- **Pie chart**

✦ *Purpose:* Quick visual understanding of patterns and outliers.

6. Summary Statistics (Descriptive Statistics)

A combined overview including:

- Count
- Mean
- Median
- Minimum
- Maximum
- Standard deviation

Purpose: Provides a quick snapshot of the dataset.